What Is AWS Cost and Usage Report (CUR)?

The AWS Cost and Usage Report (CUR) tracks AWS costs and usage at the account or organization level by usage type, operation, and product code. It lets you use Cost Categories and Cost Allocation tags to fine-tune the information.

The AWS Cost and Usage Report offers a comprehensive set of cost and usage data across AWS. It includes metadata about AWS services, credit, pricing, fees, discounts, taxes, cost categories, Savings Plans, and Reserved Instances.

You can view the Cost and Usage Report at monthly, daily, or hourly levels of granularity. Using the AWS Cost and Usage Report for both member and management accounts enables management accounts to share reports with member accounts instead of repeatedly creating reports for members.

How the AWS Cost and Usage Report works

The AWS Cost and Usage Report accumulates data over a given month and includes all current billing data for that month when you generate a report. However, each report is an estimation subject to changes according to actual usage.

Each AWS service provides usage-based billing information during different times, so updates in your report may occur during different times. AWS finalizes usage charges at the end of the month when issuing an invoice with usage charges.

Updates and refunds

AWS may update your report and apply refunds, support fees, and credits to the monthly usage. You can learn whether your bill was finalized by checking if the InvoiceId column (you can find it in the Cost and Usage Report) includes an Invoice ID for a line item. If it does, those line items are final.

AWS calculates developer, business, and enterprise support according to final usage charges, reflected on the 6th or 7th day of each month for the previous month's report. AWS applies refunds and credits according to your contract or agreement with AWS.

Related content: Read our guide to [AWS cost optimization]

AWS CUR: Pros and cons

Here are some of the key advantages of using AWS Cost and Usage Report:

- Provides details about spending on AWS—shows granular details about spending on specific Amazon resources, trends in usage and costs, and provides a better understanding of the details behind buckets of IT expenditure.
- Provides details about usage—offers usage details in very fine detail, from the types of resources used on AWS to their utilization, data transfer rates, etc. This makes it possible to understand if resources are used to their maximum potential and whether there is room for optimization.
- Easy access to granular data—when using Amazon Cost Explorer, it is difficult to gain access to the detailed, raw data provided by CUR. By using CUR, finance teams and managers can gain access to detailed information that would otherwise be inaccessible to them.

There are also disadvantages to using CUR:

- Data is complex and difficult to interpret—the report provides such detailed, specific data that can be difficult to use and understand. It is highly recommended to get assistance from an individual who has experience with these reports who can help you derive the relevant insights.
- The only way to directly receive cost and usage data—Amazon currently does not offer other options for retrieving granular data about usage of its services.

You can complement CUR with the [AWS Cost Calculator](#) to get additional insights into your Amazon costs.

Tutorial: Creating cost and usage reports

Here is how to create a Cost and Usage Report with the Billing and Cost Management console:

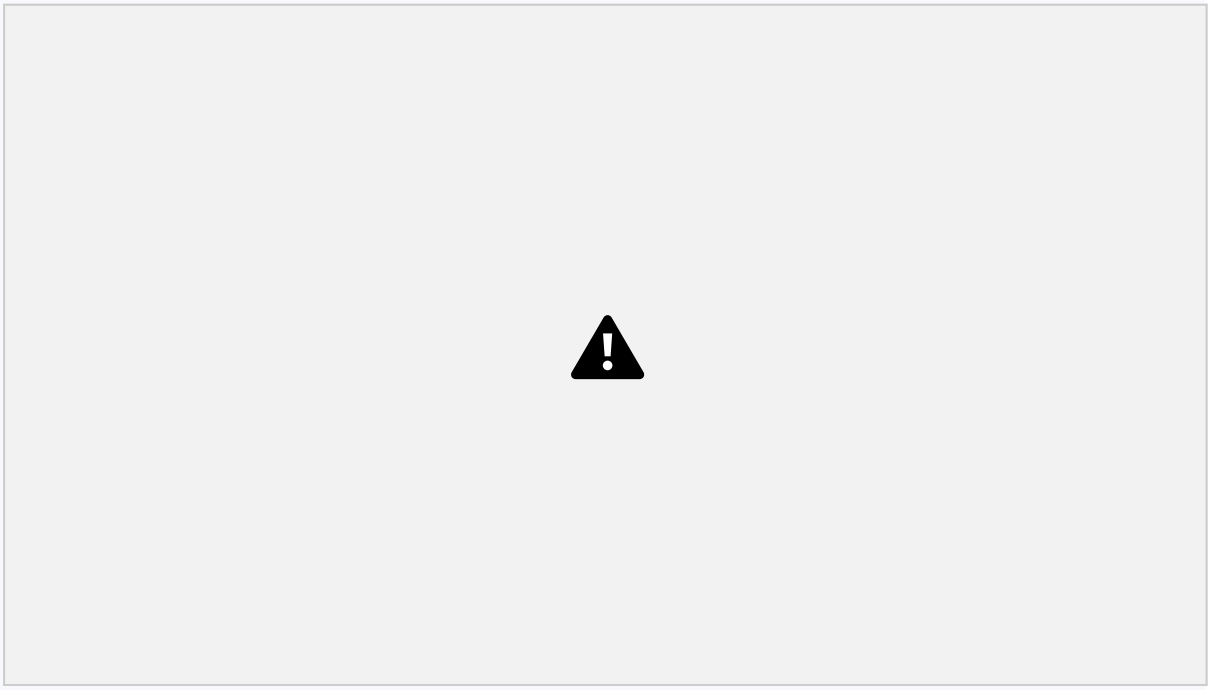


Image Source: [AWS](#)

1. Go to the Billing and Cost Management console at https://console.aws.amazon.com/billing/home#/ and sign in to your account.
2. Go to the navigation pane, select Cost & Usage Reports, and choose the Create report option.
3. In the Report name field, enter a descriptive name.
4. Go to Additional report details, and choose the Include resource IDs option to add the IDs of each resource to your report.
5. Go to Data refresh settings, specify if you want the report to refresh when AWS applies credits, support fees, or refunds when finalizing the bill, and select Next.

6. Go to S3 bucket, and select Configure.
7. Go to the Configure S3 Bucket box and choose a bucket from the drop-down (if you want an existing bucket) or specify a name region for a new bucket name. Choose Next.
8. Review the displayed bucket policy, select the confirmation option I have confirmed that this policy is correct, and select Save.
9. Go to the Report path prefix and enter a report path prefix to attach to your report's name.
10. Go to Time granularity, and select whether you want AWS to aggregate line items in the report on an hourly, daily, or monthly basis.
11. Go to Report versioning, and opt in to overwrite previous report versions with the new version or add any new version in addition to previous versions.
12. Go to Enable report data integration for, and choose whether to enable your reports to integrate with Amazon QuickSight, Amazon Redshift, or Amazon Athena.
13. Review the settings for the report, and then select Review and Complete.

Automatically optimizing AWS costs with Spot

While AWS offers Savings Plans, RIs and spot instances for reducing EC2 costs, these all have inherent challenges:

- Spot instances can be 90% less expensive than on-demand instances, however, as spare capacity, AWS can reclaim those instances with just a two minute warning, making them less than ideal for production and mission-critical workloads.
- AWS Savings Plans and RIs can deliver up to 72% cost savings, but they do create financial lock-in for 1 or 3 years and if not fully utilized can end up wasting money instead of saving it.

Spot addresses these challenges, allowing you to reliably use spot instances for production and mission-critical workloads as well as enjoy the long-term pricing of RIs without the risks of long-term commitment.

Key features of Spot's cloud financial management suite include:

- Predictive rebalancing—identifies spot instance interruptions up to an hour in advance, allowing for graceful draining and workload placement on new instances, whether spot, reserved or on-demand.
- Advanced auto scaling—simplifies the process of defining scaling policies, identifying peak times, automatically scaling to ensure the right capacity in advance.
- Optimized cost and performance—keeps your cluster running at the best possible performance while using the optimal mix of on-demand, spot and reserved capacity.
- Enterprise-grade SLAs—constantly monitors and predicts spot instance behavior, capacity trends, pricing, and interruption rates. Acts in advance to add capacity whenever there is a risk of interruption.
- Serverless containers—allows you to run your Kubernetes and container workloads on fully utilized and highly available compute infrastructure while leveraging spot instances, Savings Plans and RIs for extreme cost savings.
- Intelligent and flexible utilization of AWS Savings Plans and RIs—ensures that whenever there are unused reserved capacity resources, these will be used before spinning up new spot instances, driving maximum cost-efficiency. Additionally, RIs and Savings Plans are fully managed from planning and procurement to offloading unused capacity when no longer needed, so your long-term cloud commitments always generate maximum savings.
- Visibility and recommendations—lets you visualize all your cloud spend with the ability to drill-down based on the broadest range of criteria from tags, accounts, services to namespaces, annotations, labels, and more for

containerized workloads as well as receive cost reduction recommendations that can be implemented in a few clicks.

What is AWS cost optimization?

Amazon Web Services (AWS) is the world's leading cloud computing platform, providing hundreds of services covering compute, storage, networking, and platform as a service (PaaS) offerings such as managed databases and container orchestration.

Optimizing AWS costs involves monitoring and analyzing your usage, identifying opportunities to reduce costs, and implementing recommended actions.

Most Amazon services are based on compute resources provided by the Elastic Compute Cloud (EC2), so the cost of EC2 instances is probably a central factor in your AWS costs. Many optimization strategies involve reducing the cost of EC2 instances—for example by switching instances from on-demand pricing to lower-cost spot instances or reserved instances, or applying savings plans which can reduce costs across your Amazon account.

This is part of an extensive series of guides about FinOps.

Why is AWS so expensive?

The AWS cloud provides over 200 services. Cloud resources are dynamic and cost can be unpredictable and difficult to manage.

Here are the main causes of waste and high costs on AWS:

- Compute instances on services like Amazon EC2 are under-utilized, meaning you are paying for instances you don't actually need.
- Unused EBS volumes, snapshots, load balancers, or other resources are not in use but still incurring costs.
- Spot instances or reserved instances are not used when they are applicable – these types of instances can provide discounts of 50-90%.
- Savings Plans are not used – these can help you save on compute costs by committing to a minimum total spend on AWS.

- Auto scaling is not implemented or not optimal – for example, as demand increases, you scale up too much (adding redundant resources).

Pillars of AWS cost optimization

Here are five cost optimization pillars you can apply to most environments:

1. Right size – the resources you provision must match your requirements and needs. For example, you should provision CPU, storage, network throughput, and memory for computing.
2. Increase elasticity – traditionally, IT hardware is rarely turned off. The cloud model enables you to optimize costs to meet dynamic needs, turning resources off when you no longer need them.
3. Leverage the right pricing model – AWS offers various pricing models, including on-demand pricing, Spot Instances, and Reserved Instances. The pricing model you choose should allow you to optimize costs according to your workload's needs. For example, reserved instances are ideal for predictable workloads.
4. Optimize storage – AWS offers several storage tiers, each providing different performance at different costs. You can optimize storage and maintain the required performance and availability by identifying the appropriate destination for specific data types. For example, for lower performance requirements, you can use Amazon EBS Throughput Optimized HDD (st1), which costs less than General Purpose SSD (gp2).
5. Measure, monitor, and improve – since cloud environments are dynamic, you need to establish measurements and monitor for accurate visibility and continuous costs optimization. You can set this up by defining and enforcing cost allocation tagging. You should define metrics, set specific targets, and review the information regularly. Use training, visualization, and incentives to help teams architect for cost, and assign optimization responsibilities to certain individuals or teams.

3 AWS cost optimization tools

Here are several AWS cost optimization tools available free from Amazon:

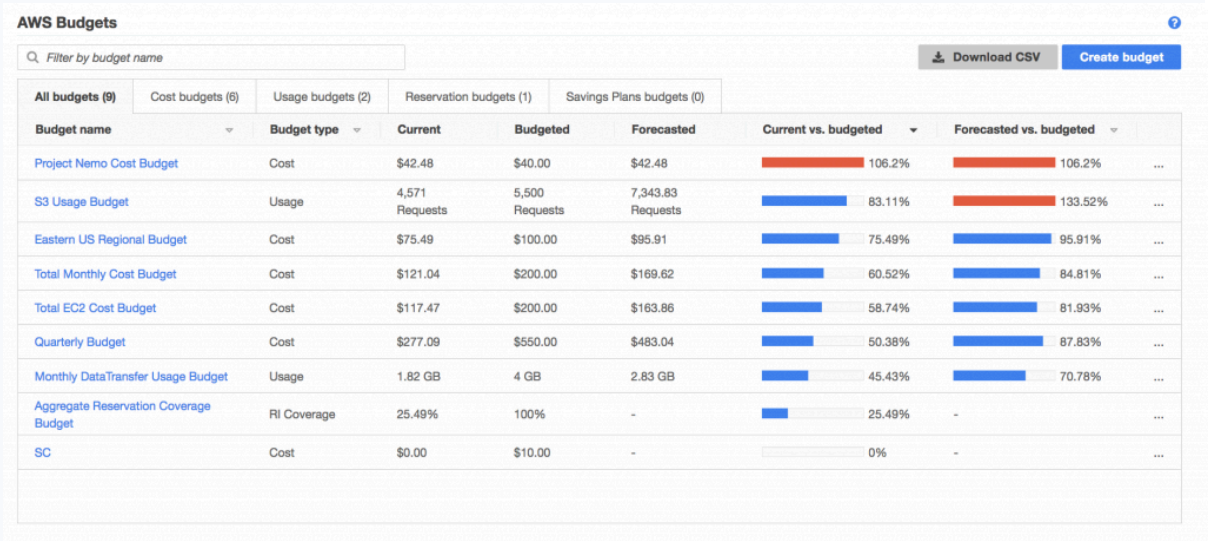
To learn other ways to reduce your cloud costs, see our detailed guide to AWS cost savings

1. AWS Cost Explorer

The Cost Explorer interface enables you to view costs, usage and return on investment (ROI) for AWS services. The interface displays data for the past 13 months and can help you forecast your future spend. You can use the interface to create customized views, which can help further analyze your AWS costs and also identify certain areas for improvement. Additionally, the AWS Cost Explorer provides an API, which enables you to access data through your existing analytics tools.

Learn more in [our detailed guide to the AWS Cost Explorer]

2. AWS budgets



AWS Budgets can help you set and enforce budgets for each AWS service. When budgets are exceeded or reached, you can receive messages or emails from the Simple Notification Service (SNS). You can define an overall cost budget or connect a budget to certain data points, including data usage or the number of instances. The tool provides dashboard views similar to those generated by the Cost Explorer, displaying how each service is used compared to its budget.

Learn more in our detailed guide to [AWS Budgets]

3. AWS pricing calculator

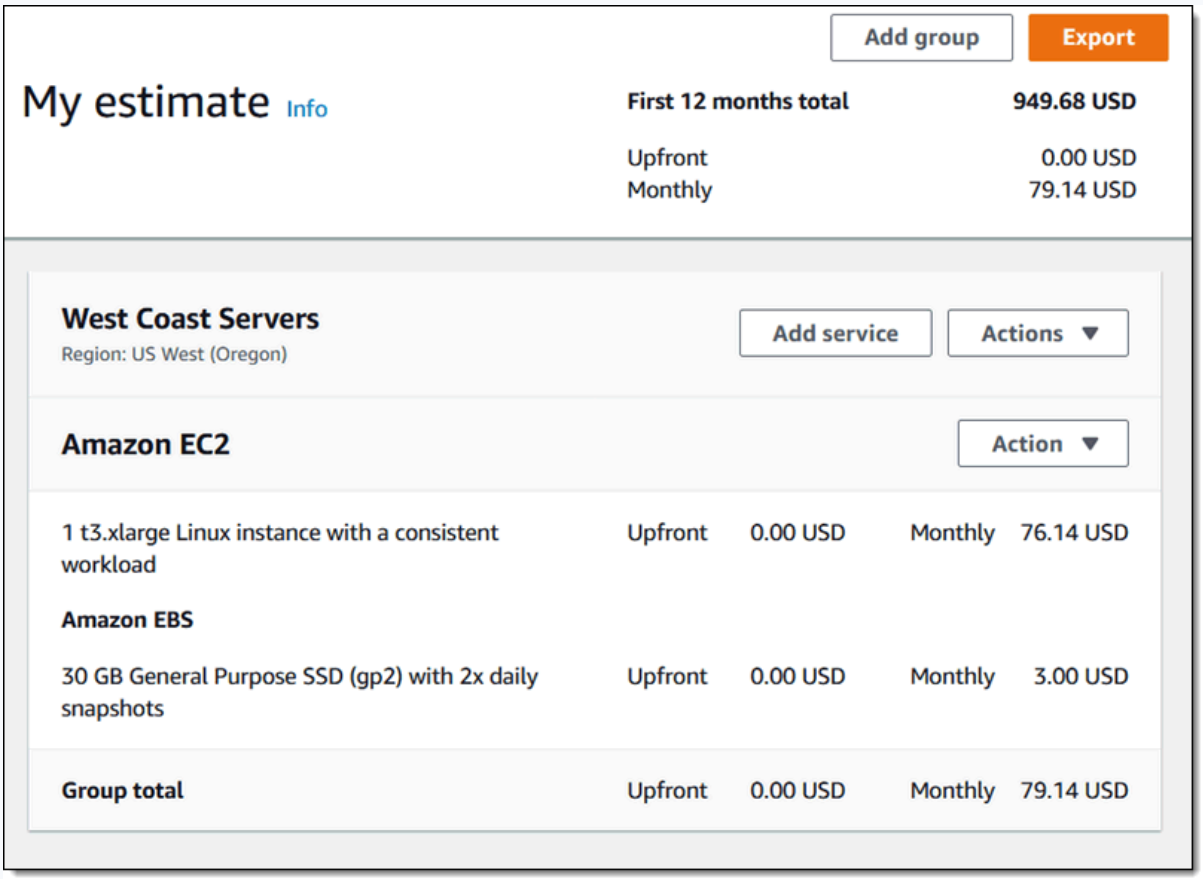


Image Source: [AWS]

AWS Pricing Calculator enables you to estimate the cost of use cases on AWS. It lets you generate monthly cost estimates for all regions supported by a certain service. You can model a solution before building it, explore the pricing points and calculations of your estimate, as well as find available instance types and contract terms that meet your

requirements. This can help you make informed decisions, plan your AWS cost and usage, as well as estimate the costs of setting up a new set of instances and services.

Read more in our guide to AWS Pricing Calculator

12 AWS cost optimization best practices

The following best practices can help you optimize and reduce your AWS costs.

1. Choose the appropriate AWS region

To use the AWS Management Console, CLI or SDK, you must first choose a region. Most users base this decision on distance. However, the AWS region you choose also affects costs, latency and availability. Here are several aspects to consider when choosing the right AWS region for your project:

    - Costs—each AWS region is priced differently. You can view official on-demand pricing on the AWS site, but make sure to use the cost calculator to estimate your costs when using a certain region.
    - Latency—regions with smaller latency can help make the application more accessible to a certain group of users.
    - Service availability—not all AWS services are made available to all regions. Before choosing a particular region, make sure the target AWS service is available in this region.
    - Availability—using multiple AWS regions can improve availability and enable you to create a separate disaster recovery site.
- Data sovereignty—when you store data in a specific geographical location, your organization must comply with that location's legal requirements for the data. This is an important consideration for locating your sensitive data.

When accounting for all above aspects, you can prioritize them to choose the most important aspect and let it guide you as you choose a region.

2. Create schedules to turn off unused instances

When optimizing your AWS costs, you must pay attention to unused instances and shut them down. Here are several practices to consider:

- Shut down unused instances at the end of a working day or during weekends and vacations.
- When optimizing non-production instances, you should prepare on and off hours in advance.
- Evaluate usage metrics and then decide when instances are frequently used. You can then implement more accurate schedules. Alternatively, you can apply an always-stopped schedule, which you can disrupt when providing access to these instances.
- Determine whether you are paying for EBS quantities as well as other relevant elements while your instances are not in use.

Read more in our guide to AWS Cost Anomaly Detection

3. Identifying under-utilized Amazon EC2 instances

The AWS Cost Explorer helps you visualize and manage the cost of Amazon services. The tool provides a Resource Optimization report, which shows idle or under-utilized EC2 instances. You can lower your costs by stopping or scaling down these instances.

Here are three tools to help you to stop wasting money on low utilization EC2 instances:

- AWS Instance Scheduler – a pre-integrated Amazon solution you can use to stop instances automatically on a predetermined schedule, for example outside business hours.
- AWS Operations Conductor – automatically resizes EC2 instances based on recommendations provided by Cost Explorer.
- AWS Compute Optimizer – provides recommendations about the most appropriate instance type for each workload. This goes beyond scaling down within a group of instances. It offers recommendations for downsizing instances across groups, as well as recommendations for upsizing to eliminate performance bottlenecks, and recommendations for EC2 instances in an Auto Scaling group.

4. Reducing EC2 costs with EC2 spot instances

Amazon EC2 spot instances are an important option for reducing AWS costs. Spot instances provide savings of up to 90% compared to normal on-demand prices.

Spot instances are Amazon's way to sell off spare EC2 capacity. They can significantly reduce Amazon EC2 costs by allowing you to request the same EC2 instances, when they are in low demand, at a significantly discounted price.

The downside of spot instances is that they are unreliable—if Amazon needs to reclaim the capacity for on-demand or reserved users, it terminates the instance with a two-minute warning. Amazon has recently introduced "rebalancing signals" which may provide earlier warning of spot instance termination, but this is not guaranteed. Another option to improve reliability is to run spot instances in an Auto Scaling Group (ASG) together with regular on-demand instances. This way at least some of the capacity in the group is guaranteed to remain available.

You can overcome these challenges and use spot instances even for mission critical or production workloads that cannot tolerate interruption with cloud optimization tools like Elastigroup from Spot. Elastigroup provides AI-driven prediction of spot instance interruptions, and can perform automated rebalancing of workloads, to eliminate the "surprise factor" and risk of spot instance termination.

5. Optimizing EC2 Auto Scaling Groups (ASG) configuration

An ASG is a collection of Amazon EC2 instances, treated as a logical group for automatic scaling and management purposes. ASGs can take advantage of Amazon EC2 Auto Scaling features, such as health checks and custom scaling policies based on application metrics or preset schedules. You can dynamically add or remove EC2 instances from an ASG based on predetermined rules or dynamically, in response to application loads.

ASGs let you scale your EC2 fleets up or down as needed to conserve costs. You can view scaling activity by using the describe-scaling-activity CLI command, or the Auto Scaling Console. By optimizing scaling policies, you can reduce costs both when scaling up and down:

- When scaling up—try to add instances less aggressively, monitoring to see that application performance is not affected
- When scaling down—try to reduce instances to the minimum necessary to maintain current application loads

Here too, a tool like Elastigroup can help by exactly matching scaling events to application requirements, ensuring there are no wide margins that waste capacity. This can also save your team time, because there is no need to constantly monitor and fine tune auto scaling policies.

Learn more about using Elastigroup to optimize EC2 auto scaling

6. Using or selling under-utilized reserved instances

An Amazon Reserved Instance (RI) lets you commit to using an instance for periods of one or three years, providing a discount of up to 72%. There are several options you need to consider when committing to reserved instances:

- Standard or Convertible—you can resell standard RIs on the AWS RI Marketplace if you no longer need them, but you cannot change the type of instance group. You cannot resell convertible RIs, but you can change them to any instance type or family.
- Regional or Zonal—regional RIs let you move the instance to a different zone, and also allow you to change to an equivalent instance size within the same family, but do not guarantee capacity. Zonal RIs guarantee capacity, without letting you change zone or instance type.
- Payment options—you can decide whether to pay upfront for the commitment period, pay some of the amount upfront, or pay everything on an ongoing basis. Upfront payment grants larger discounts.
- Service support—you can now use RIs for EC2, RDS, Redshift, ElastiCache and DynamoDB.

Because RIs represent a long-term commitment, you may find yourself with unused reserved instances. This is why initial planning is important:

- If you are sure you'll need RI capacity during the commitment period, opt for Convertible instances, so you can reuse the instance for other workloads if necessary.
- If there is a chance you won't need some of the RIs for the entire commitment period, opt for Standard instances and sell them on the marketplace if needed.

Here too, automated optimization tools can be useful. Eco from Spot can help you automatically adjust your portfolio, to offload unused reserved instances, or relocate workloads from expensive on-demand instances.

Learn how to use Eco from Spot to optimize the use of EC2 reserved instances

7. Leveraging compute savings plans to reduce compute costs

The Savings Plan is a flexible pricing model that lets users use EC2, Lambda, and Fargate at a lower cost, by committing to a continuous usage, measured in terms of USD per hour, for a period of 1 or 3 years. For example, a one-year Savings Plan with no upfront payment plan grants a discount of up to 54%.

Savings Plans apply to compute instances regardless of size, Auto Scaling Group, Availability Zone, or region. You can use AWS Cost Explorer to select the right options for your plan, based on an analysis of recent utilization.

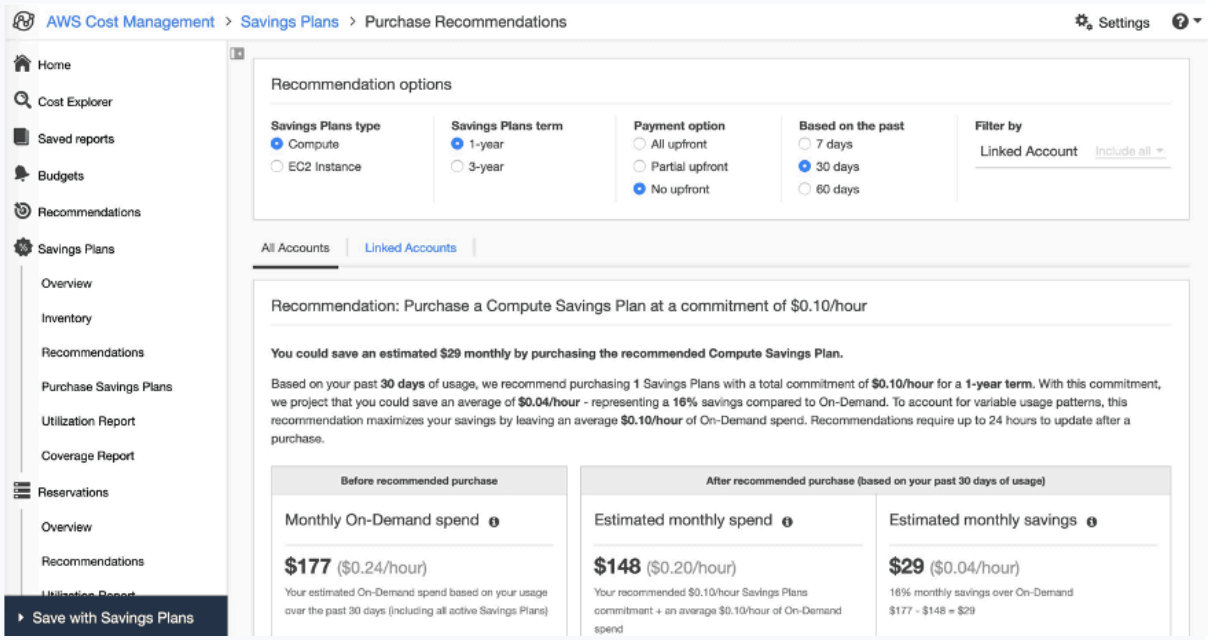


Image Source: [AWS]

While Cost Explorer provides useful recommendations, your usage of Amazon services is dynamic, and there is a need to continuously manage and optimize your commitments. A cloud optimization tool like Eco constantly evaluates cloud usage and manages Savings Plans and reserved instance lifecycles automatically, to ensure you make the best use of long term commitments and achieve maximal discounts.

Learn how to use Eco from Spot to [optimize the use of EC2 savings plans]

8. Monitor the use of storage and delete unused EBS volumes

Monitoring can help you track actual usage and identify pricing patterns. AWS lets you monitor S3 usage via the S3 Analytics tool, which evaluates storage access patterns on a specific object dataset for a duration of 30 days or more.

The S3 Analytics tool offers recommendations that can help you leverage S3 Infrequently Accessed (S3 IA), AWS Glacier, or AWS Glacier Deep Archive for cost reduction.

You can then use lifecycle policies to automate the transfer of objects into a lower cost storage tier. Alternatively, you can use S3 Intelligent-Tiering for automatic analysis and transfer of objects with unknown or dynamic usage duration into a relevant storage tier.

In addition, a great way to save storage is to delete unused Elastic Block Storage (EBS) volumes – managed disk drives you can attach to EC2 instances. Even after the EC2 instance shuts down, EBS volumes can continue operating and incur costs.

Make sure teams are aware that when they use EBS volumes, they should select the option to Delete on termination. This will ensure that the EBS volume is deleted when the EC2 instance is terminated. If this was not done, look for EBS volumes that are marked as available—you can do this via Amazon CloudWatch or AWS Trusted Advisor, and clean them up automatically using a Lambda function.

9. Identifying and deleting orphaned snapshots

When you terminate an EC2 instance, the associated EBS volume is also deleted by default. However, it's easy to forget that snapshots you created as backups of those EBS volumes are still in S3, and you're still paying an ongoing monthly fee to store them.

EBS backups are usually incremental, so each additional snapshot takes up limited storage space. However, if you perform frequent snapshots and have a high retention period, even this small incremental addition can add up over time.

Because most snapshots use data from the initial snapshot of the entire EBS volume, it is important to track down and delete the initial snapshot, if it is not needed. This can save much more storage space than deleting numerous incremental snapshots.

It is best to set up automated lifecycle management of EBS snapshots, via Amazon Data Lifecycle Manager, to ensure that you do not maintain snapshots for longer than needed.

10. Deleting idle load balancers and optimizing bandwidth use

Check your Elastic Load Balancing configuration to see which load balancers are not currently used. Every load balancer incurs ongoing costs. If your load balancer doesn't have any backend instances associated with it, or network traffic is very low, it is not being used effectively and is wasting resources.

You can use AWS Trusted Advisor to identify load balancers with a low number of requests (a good rule of thumb is less than 100 requests in the last 7 days). Reduce costs by removing idle load balancers—you can track overall data transfer costs with Cost Explorer.

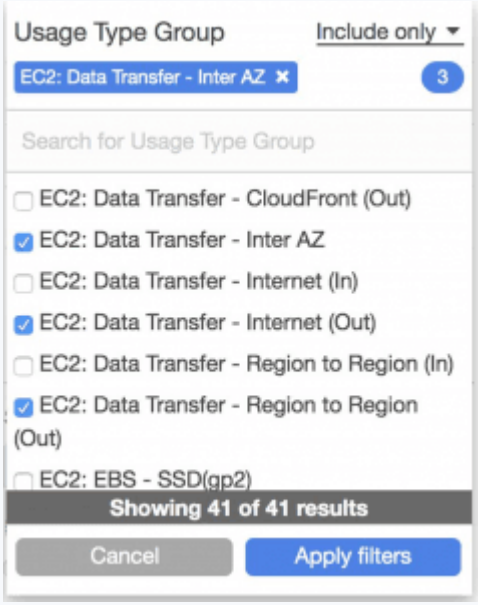


Image Source: AWS

Another way to reduce costs, if you have high data transfer costs from EC2 to the public web, is to use Amazon CloudFront. CloudFront is a Content Delivery Network (CDN), which lets you cache web content in multiple edge locations around the world. This can significantly reduce the bandwidth required to serve spikes in traffic.

11. Charging back Amazon costs to internal users with enterprise billing console

AWS Enterprise Billing Console is a new service that lets you manage chargebacks—billing Amazon services to units in your organization, or to third parties. Within an organization, this allows you to create accountability by billing each department or business unit, according to the cost of services they actually use.

The Enterprise Billing Console lets you allocate your costs across accounts, and use the concept of billing groups. Billing groups let you apply customized pricing plans to each department or business unit. Set up Cost and Usage (CUR) reports for each of your billing groups, and perform margin analysis to calculate savings for each group as a result of cost optimizations.

Another feature of the console is dashboard graphs, which offer default views of your overall spending, as well as month-to-date spend and service consumption by each group.

Spot Eco can be used by enterprises looking to easily manage cost, internal billing and showback for various business units.

Related content: Read our guide to AWS cost management

12. Automatically optimizing cloud costs with Spot

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What are AWS Budgets?

AWS Budgets enables you to create custom budgets to track AWS costs and usage. Once you set this up, you can get email alerts or SNS notifications. You can set alerts to notify you:

- When forecasted or actual usage and cost exceed the budget threshold.
- When actual Reserved Instances and Savings Plans coverage or utilization drops below the desired threshold.

AWS Budget Actions enable you to configure actions in response to usage and cost status. If a threshold exceeds, actions are executed—automatically or with human approval—to reduce over-spending.

AWS Budgets integrates with various AWS services, including:

- AWS Cost Explorer—enables you to easily view and analyze cost and usage drivers.
- AWS Chatbot—delivers Budget alerts in a designated Slack channel or Amazon Chime room.
- AWS Service Catalog—lets you track costs on approved AWS products and portfolios.

There are several budget types you can create in AWS Budgets:

- Cost—set your expected expenditure on each service.
- Usage—set your expected usage of one or several services.
- RI utilization—set a threshold for reserved instance utilization to track underused or unused RIs.
- RI coverage—set an RI coverage threshold to detect when you have fewer instance horses covered by reserved instances.
- Savings plan utilization—set a threshold to track underused savings plans.
- Savings plan coverage—set a threshold to track the coverage of your savings plans and identify underutilized available instance usage.

You can configure alerts to notify you when you fall below, exceed, or are about to exceed your specified budgets.

There are several ways to set your usage or costs budget:

- Fixed budget—monitor a fixed budgeted amount over all periods.
- Planned budget—monitor different amounts based on quarterly or monthly budgets. You can plan your budget for up to a year (4 quarters/12 months), after which the last planned budget will become a fixed budget.
- Auto-adjusted budget—dynamically set your budgeted amount based on usage or spending over a specified time range. The time range you choose is the baseline for auto-adjusting the budget, with AWS Budgets calculating the budget for each period based on your baseline. Choosing the right time range for your expected usage or costs is important to avoid receiving too few or too many alerts.

Related content: Read our guide to AWS cost optimization

Tutorial: Creating a budget

When you create a budget, you first need to choose your monthly target budget value. You can then go to the AWS Budgets console and create the budget based on your chosen value. The objective of a budget is to prevent you from overspending across all the AWS

services you use. For this reason, a good first option is to set the cost budget to track your overall spending.

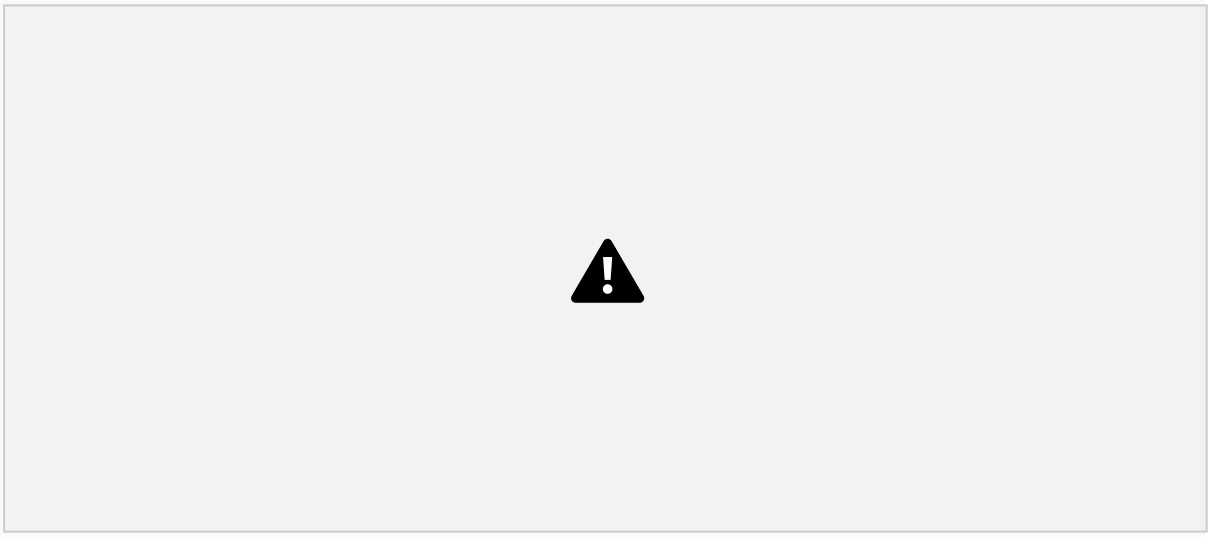

Image Source: AWS

When you set your budget amount, you should specify the following:

- Period—monthly
- Budget effective date—recurring budget
- Start month—use the default
- Choose how to budget—fixed
- Budgeted amount—enter your chosen value (ideally determined in advance)

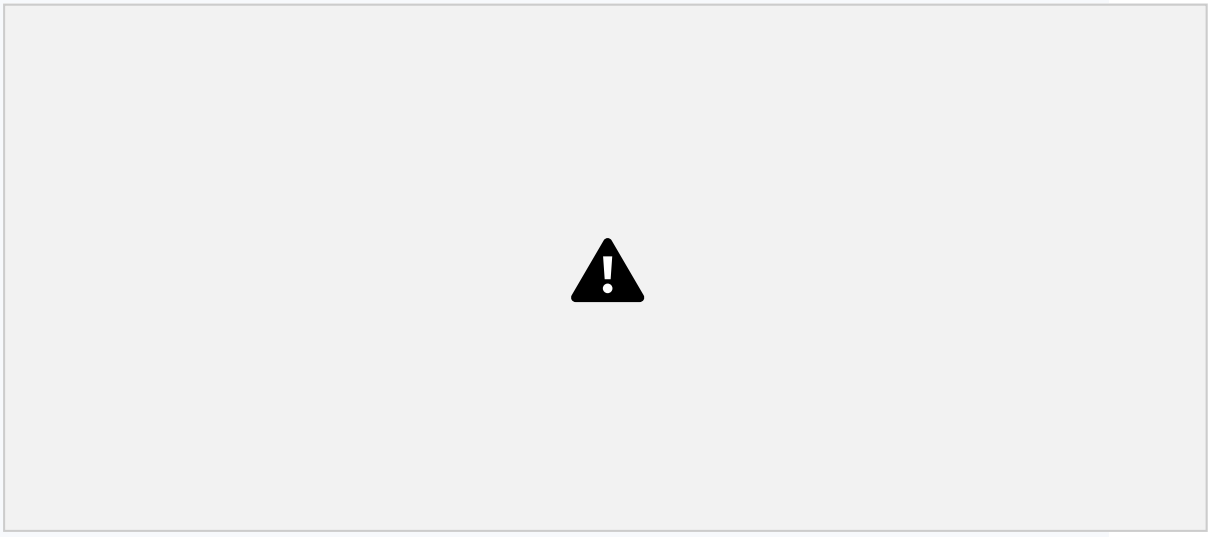

Image Source: AWS

Once you've set your budget amount, you can choose to apply a scope to the budget to create budgets for specific groups of objects, such as services, accounts, regions, or tags. In this case, the budget will serve as a runaway spending budget to catch high-level overspending, so you do not need to apply any scopes.

Generally, it is best to create separate budgets for each workload you run. Budget scopes can help you monitor aspects such as development and testing costs for new instances or services. Scopes are also useful for setting budgets for specific teams in the organization.

You can set up alerts to receive notifications when your costs reach your specified thresholds. It is important to find the right balance—you need to monitor your spending without receiving too many notifications and causing alert fatigue. A good option is to create two alert thresholds, one that sends email notifications when you reach 50% of the budgeted amount and one that notifies you when you reach 75%.

Image Source: AWS

Best practices for AWS budgets

Using managed policies

AWS Budgets cannot execute configured actions if you do not set up and assign permissions configured for users and AWS Budgets. AWS provides managed policies to help ensure your actions work as intended. Here are the two main policies:

- User policy—ensures the user can pass a role to the Budgets service.
- Budgets policy—enables the Budgets service to execute your action.

AWS recommends using identity and access management (IAM). It helps avoid updating existing IAM policies for AWS Budgets every time new functionality is included. AWS adds new capabilities to your managed policy by default.

Setting budget alerts

You can configure the system to send Budget alerts to up to ten email addresses. It is restricted to one Amazon Simple Notification Service (Amazon SNS) topic per alert. It lets you set budgets to alert against actual or forecasted values. Here is how this works:

- Actual alerts—the system sends actual alerts once per budget period when the budget first reaches a certain threshold.
- Forecast-based budget alerts—the system sends these alerts on a per-budget period basis. It may push an alert more than once during a budgeted period when the forecasted values exceed or go below. AWS requires about five weeks of usage data to create budget forecasts.

Controlling access to AWS Budgets

You can enable IAM users to create budgets in your AWS Billing and Cost Management console by providing IAM users with the following privileges:

- Viewing billing information
- Creating Amazon SNS notifications
- Creating Amazon CloudWatch alarms

AWS lets you create budgets programmatically using the Budgets API. When you configure access to this API, you need to create a unique IAM user for programmatic access. It helps define more precise access controls between the API and roles in the organization with

access to the AWS Budgets console. You can give multiple IAM users query access to the API by creating a programmatic access IAM role for each user.

Related content: Read our guide to AWS cost savings

Optimize AWS costs and budgets with Spot

While AWS offers Savings Plans, RIs and spot instances for reducing EC2 cost, these all have inherent challenges:

- Spot instances can be 90% less expensive than on-demand instances, however, as spare capacity, AWS can reclaim those instances with just a two minute warning, making them less than ideal for production and mission-critical workloads.
- AWS Savings Plans and RIs can deliver up to 72% cost savings, but they do create financial lock-in for 1 or 3 years and if not fully utilized can end up wasting money instead of saving it.

Spot addresses these challenges, allowing you to reliably use spot instances for production and mission-critical workloads as well as enjoy the long-term pricing of RIs without the risks of long-term commitment.

Key features of Spot's cloud financial management suite include:

- Predictive rebalancing—identifies spot instance interruptions up to an hour in advance, allowing for graceful draining and workload placement on new instances, whether spot, reserved or on-demand.
- Advanced auto scaling—simplifies the process of defining scaling policies, identifying peak times, automatically scaling to ensure the right capacity in advance.
- Optimized cost and performance—keeps your cluster running at the best possible performance while using the optimal mix of on-demand, spot and reserved capacity.
- Enterprise-grade SLAs—constantly monitors and predicts spot instance behavior, capacity trends, pricing, and interruption rates. Acts in advance to add capacity whenever there is a risk of interruption.

- Serverless containers—allows you to run your Kubernetes and container workloads on fully utilized and highly available compute infrastructure while leveraging spot instances, Savings Plans and RIs for extreme cost savings.
- Intelligent and flexible utilization of AWS Savings Plans and RIs—ensures that whenever there are unused reserved capacity resources, these will be used before spinning up new spot instances, driving maximum cost-efficiency. Additionally, RIs and Savings Plans are fully managed from planning and procurement to offloading unused capacity when no longer needed, so your long-term cloud commitments always generate maximum savings.
- Visibility and recommendations—lets you visualize all your cloud spend with the ability to drill-down based on the broadest range of criteria from tags, accounts, services to namespaces, annotations, labels, and more for containerized workloads as well as receive cost reduction recommendations that can be implemented in a few clicks.